

SAFETYVERSE · BY UPL

Developer Guide

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Version: 2026-09 · **Audience:** Developers & Ops

Live app: <https://upl-fire-portal.onrender.com>

Repository: <https://github.com/paramshah1903/fire-portal>

SafetyVerse — Developer Guide

A gentle, illustrated tour of the codebase for anyone joining the project.

Read time: 20 minutes. By the end you'll know how the whole system fits together and where to look when you need to change something.

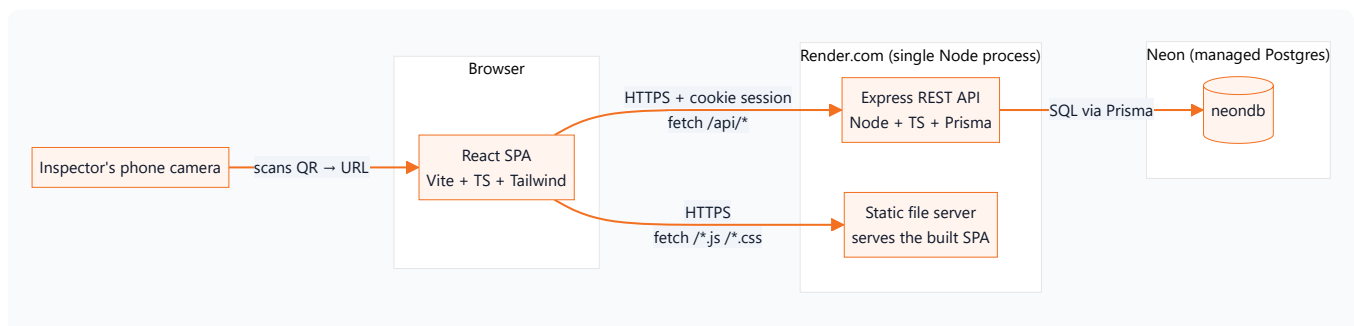
1. What this application does

SafetyVerse (internally the UPL Fire Equipment Maintenance Portal) is a role-based web app that helps a company track every piece of fire safety equipment (extinguishers, hydrants, hoses, alarms, etc.) across multiple plant units, and captures periodic inspections as immutable digital records.

The typical workflow:

1. An admin loads the equipment master (each item has a unique QR code)
2. An admin defines a **checklist template** (questions per equipment type)
3. An inspector scans a QR code with their phone → the app opens on the matching equipment → they fill the checklist → submit
4. Failed answers automatically raise **corrective actions**
5. Every submitted inspection is stored with a unique number (`INS-2026-000042`) and shows up in reports + PDF exports

2. The 10,000-foot view



Deployment topology in words:

- One **web service** on Render's free tier runs a single Node process
- That process serves both `/api/*` (Express routes) and `/*` (the built React SPA static files) from the same origin
- The database is a managed Postgres on **Neon** (also free tier)

- Photos uploaded during inspections live on the Render container's local disk (ephemeral on the free tier — see §8 for how to fix that for a real rollout)

3. Repository layout

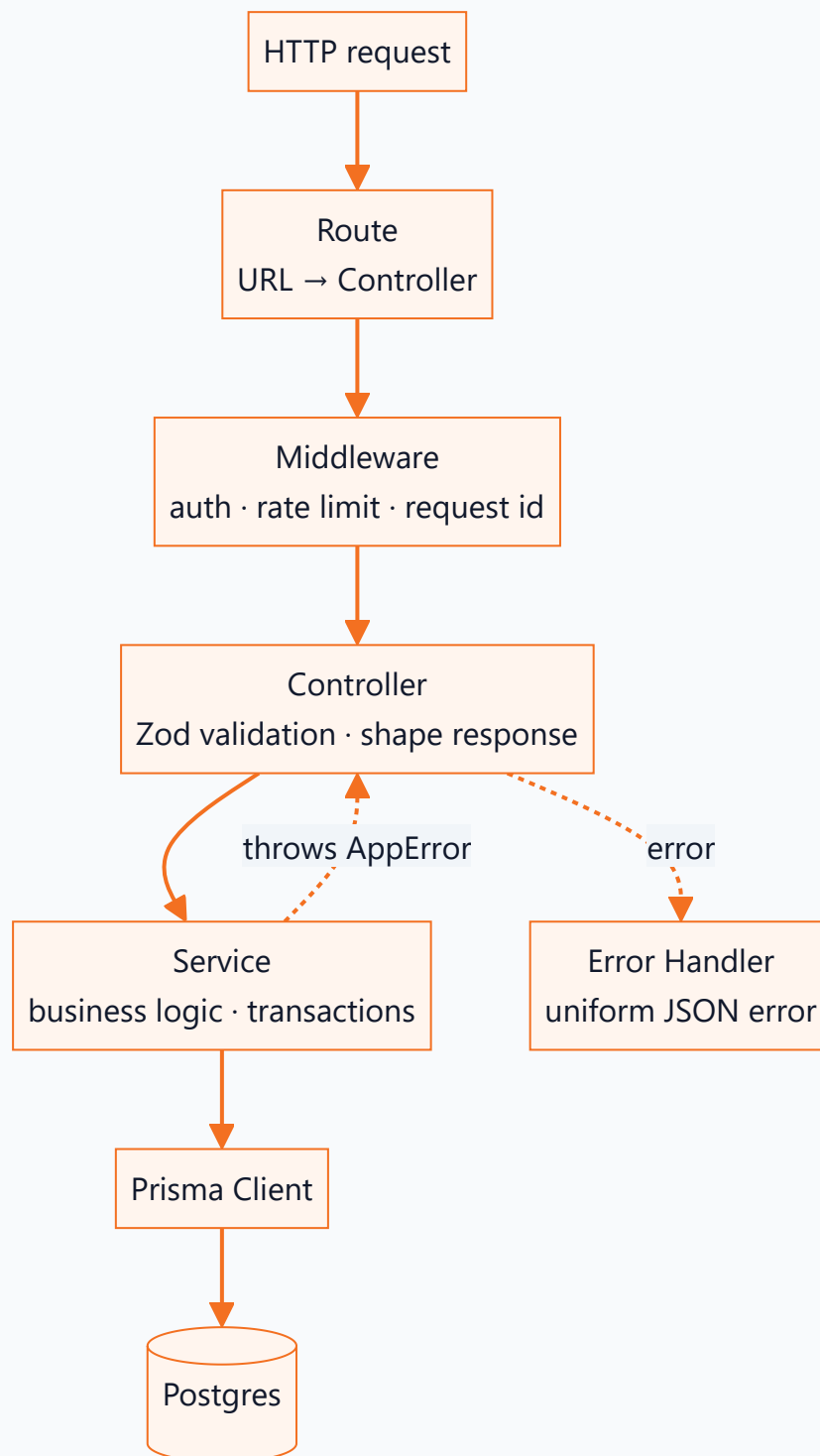
```

upl_project/
├── frontend/                # React + Vite SPA
│   ├── public/              # static assets (logo.jpeg, favicon)
│   ├── src/
│   │   ├── auth/            # AuthContext + ProtectedRoute
│   │   ├── components/      # ui.tsx, Modal, ErrorBoundary, DateRangeFilter
│   │   ├── layout/          # AppLayout (top bar + sidebar)
│   │   ├── lib/             # api client + one file per API area
│   │   ├── pages/           # one folder-less .tsx per route
│   │   │   └── reports/     # all 7 report pages
│   │   ├── theme/           # ThemeContext (light/dark toggle)
│   │   ├── App.tsx          # React Router routes
│   │   └── main.tsx         # ReactDOM entry
│   ├── index.html
│   ├── tailwind.config.js   # UPL orange brand palette
│   └── vite.config.ts
├── backend/                 # Express + Prisma API
│   ├── prisma/
│   │   ├── schema.prisma    # data model (14 tables)
│   │   └── seed.ts          # idempotent demo data
│   ├── src/
│   │   ├── config/env.ts    # typed env-var loader
│   │   ├── controllers/     # one file per resource - thin HTTP layer
│   │   ├── services/        # one file per resource - business logic
│   │   ├── routes/          # one file per resource - URL → controller
│   │   ├── middleware/      # auth, rate limit, error handler, request id
│   │   ├── lib/             # cross-cutting: rbac, prisma, errors, exporters
│   │   ├── app.ts           # Express app setup
│   │   └── server.ts        # HTTP listener
│   └── package.json
├── docs/                    # you are here
├── logo.jpeg                # UPL brand image (also copied to frontend/public/)
├── render.yaml              # Render.com deployment blueprint
├── .gitignore
└── README.md

```

4. The two big layers

4.1 Backend — clean 4-layer stack

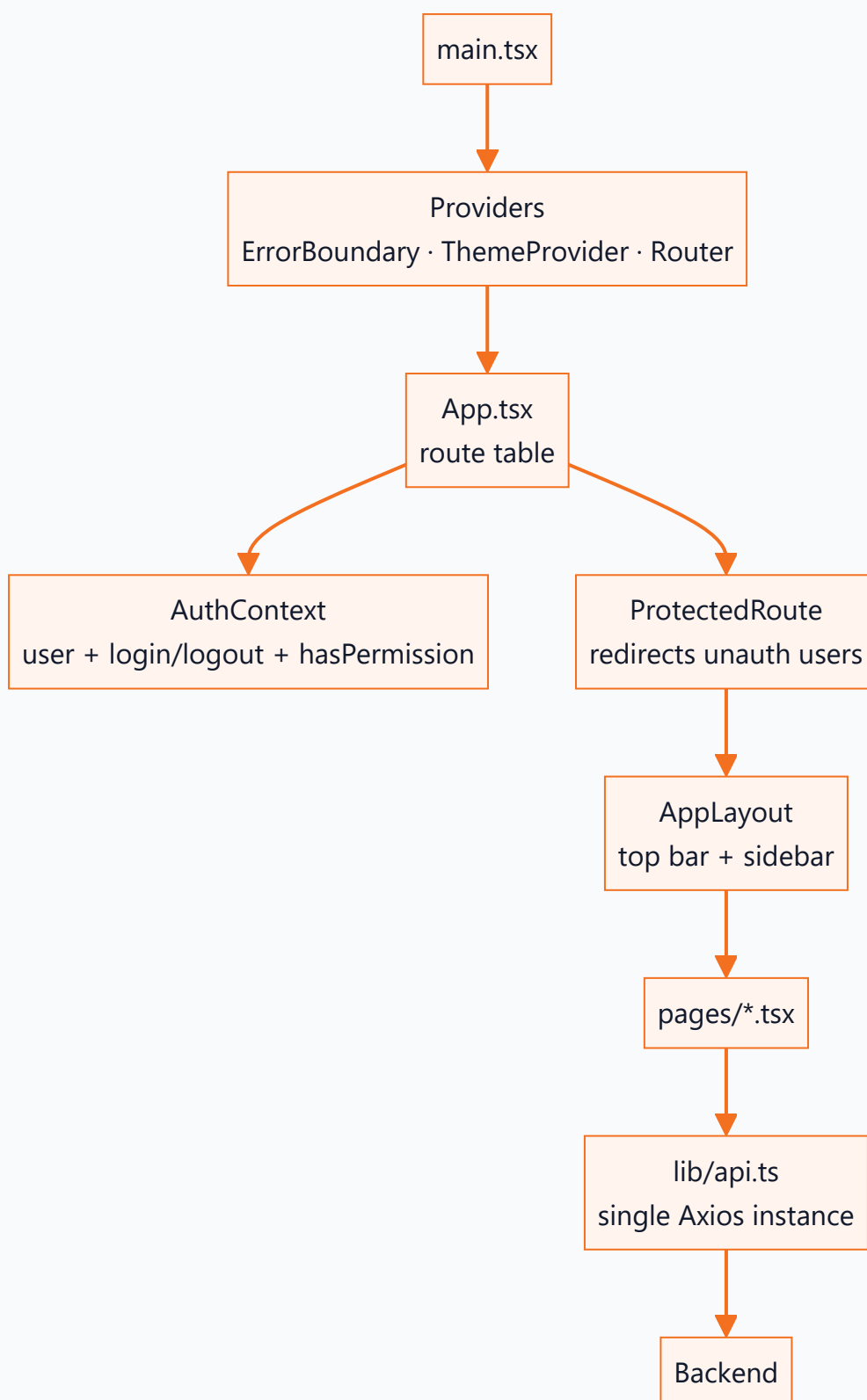


Rule of thumb — where does new code go?

Kind of code	Layer	Example
"Bind /foo to the controller"	<code>routes/foo.ts</code>	<code>router.get('/', ctrl.list)</code>
"Validate input and shape output"	<code>controllers/fooController.ts</code>	Zod schema + <code>res.json({ ... })</code>
"Compute something / talk to DB"	<code>services/fooService.ts</code>	<code>prisma.foo.findMany(...)</code>
"Run before every request"	<code>middleware/xxx.ts</code>	attach user, rate limit
"Shared helper / no I/O"	<code>lib/xxx.ts</code>	date range parsing, RBAC keys

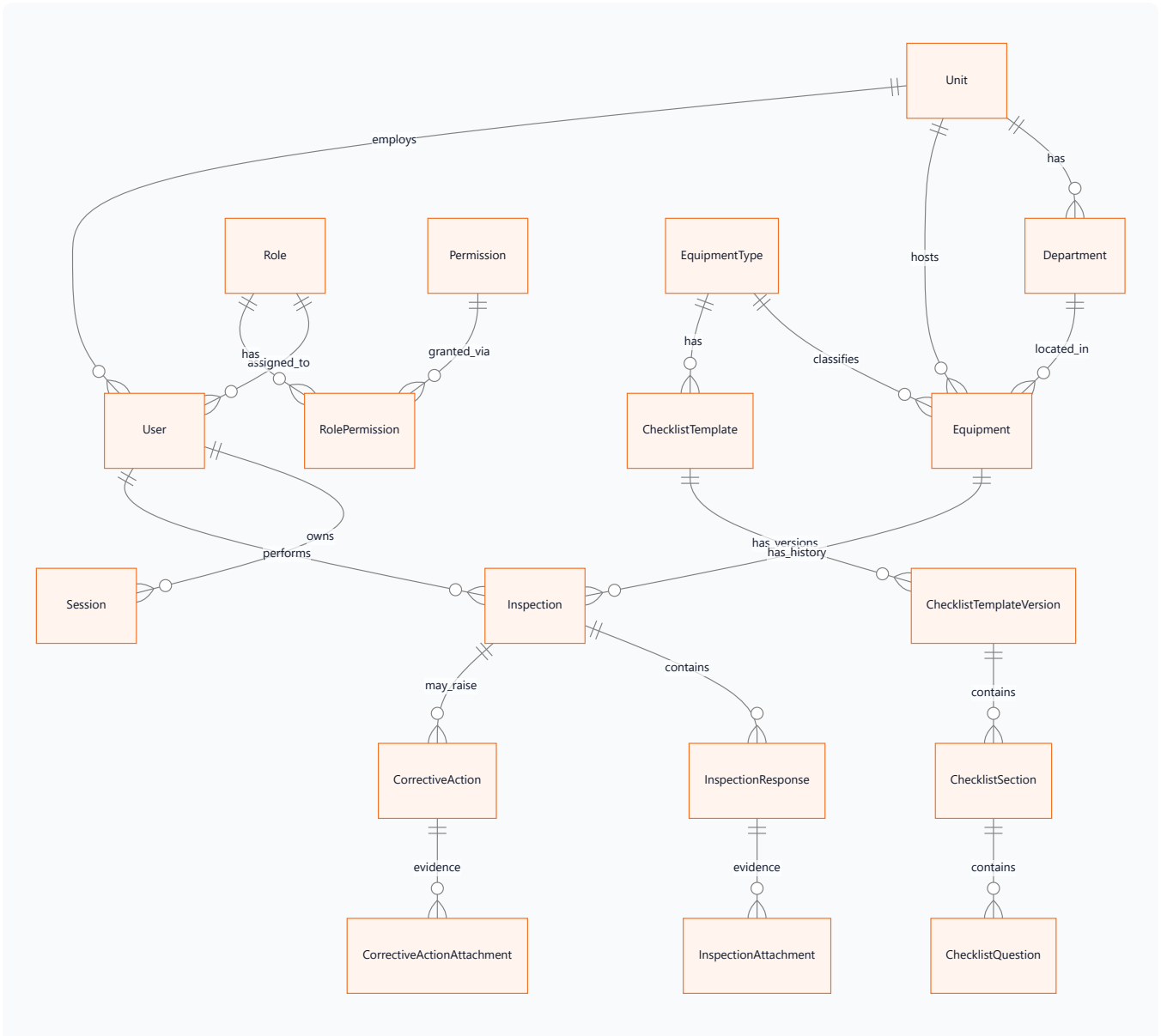
Never call Prisma directly from a controller. **Never** call `res.json` from a service. Keeps concerns clean.

4.2 Frontend — routes, hooks, one big Axios client



Every API call goes through **one Axios instance** in `lib/api.ts` so cookies, base URL, timeout, and error normalisation are consistent. Feature-specific wrappers live in `lib/apiInspections.ts`, `lib/apiEquipment.ts`, etc.

5. Data model at a glance



Fourteen tables total. The full schema is in `backend/prisma/schema.prisma` and each model has a comment explaining what it's for.

Key design decisions:

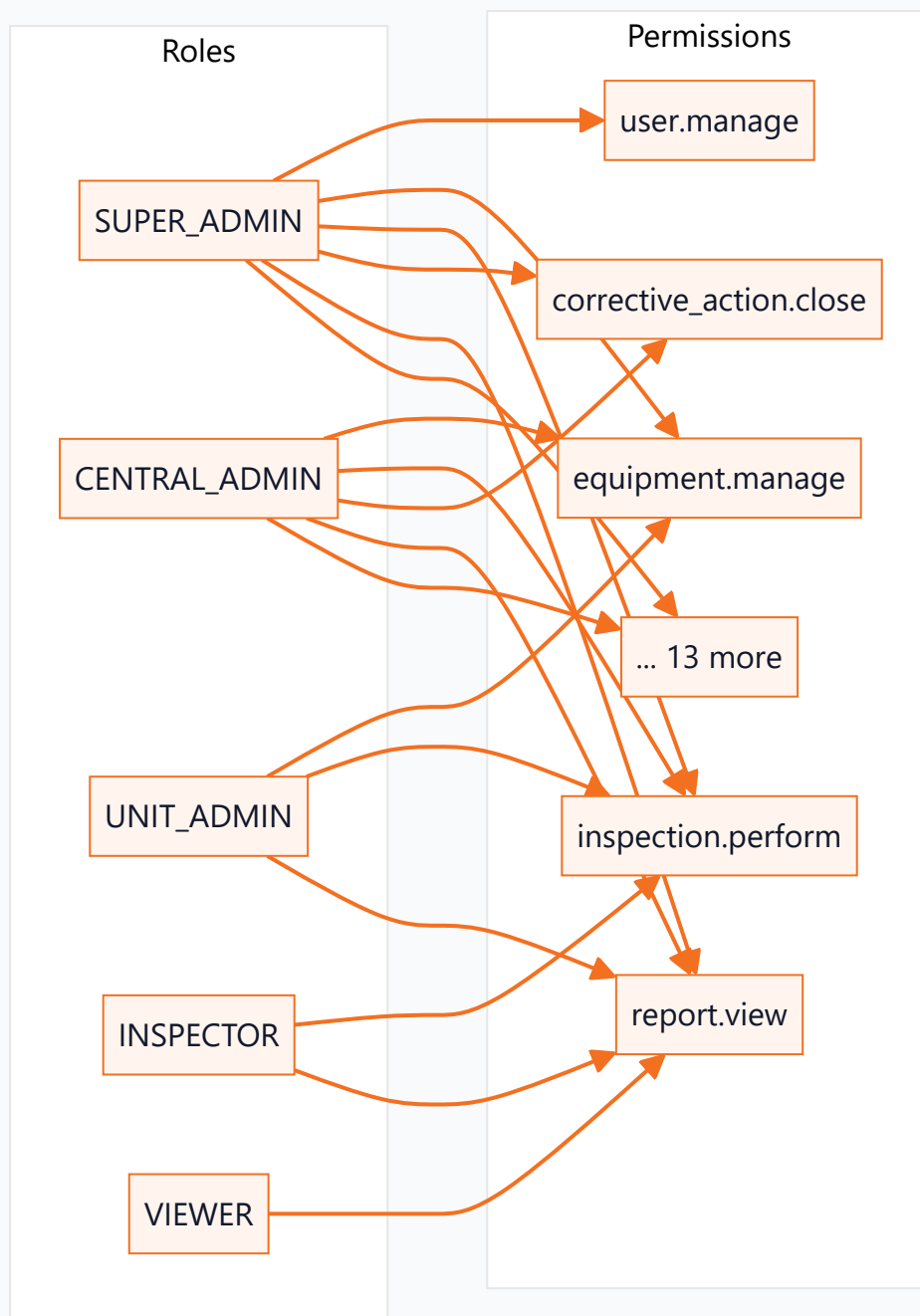
- **Versioned checklists:** editing a template creates a new version. Historical inspections keep the exact version they were captured against, so a template edit never mutates history.
- **Snapshotted responses:** each `InspectionResponse` stores a copy of the question text + type at submission time — reports and PDFs render correctly even years later.
- **Unique period key** (`YYYY-MM`) enforces "one inspection per equipment per month" at the DB level with a unique index.
- **Inspection numbers** (`INS-2026-000042`) generated inside the submit transaction — monotonic per year, unique index.

| 6. Auth and RBAC in one page

6.1 Sessions

- User logs in with username + password → backend verifies bcrypt hash
- Backend creates a `Session` row → returns an HTTP-only **signed cookie** containing the session id
- Cookie is `secure` in production (HTTPS only), `sameSite: lax`
- On every subsequent request, middleware reads the cookie, looks up the session, hydrates `req.user` (or `null` if invalid / expired)
- Sessions have **sliding expiry** (default 8h idle, 30d absolute cap)

6.2 RBAC — 5 roles, 18 permissions



- **Every backend service** re-checks permissions AND scope (unit isolation for UNIT_ADMIN / INSPECTOR)
- **Frontend hides** nav items and controls the user can't access, but the backend is the source of truth

Full role/permission matrix is in [RBAC.md](#).

7. End-to-end request walk-through

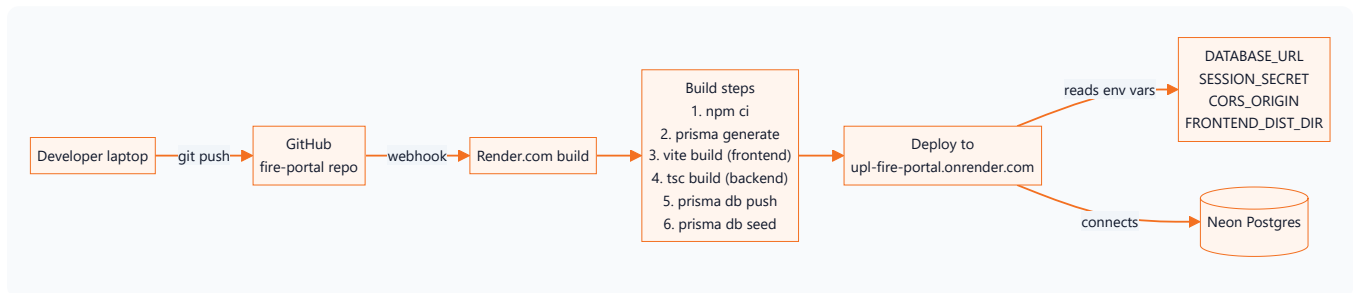
An inspector opens a QR code on their phone → we submit an inspection.



Syntax error in text

mermaid version 10.9.0

8. Deployment (production)



render.yaml at the repo root is the source of truth for build + start commands and env-var declarations. Every push to **main** on GitHub triggers a rebuild automatically.

Ephemeral disk caveat

Render's free tier wipes the container's filesystem on every restart (~15 min idle spin-down or a fresh deploy). Structured data (users, inspections, corrective actions) is safe in Neon. **Uploaded photos are not.** For a real rollout, switch photo uploads to object storage (Cloudflare R2 or S3) — see §10.

9. Local development

```
# 1. clone + install
git clone https://github.com/paramshah1903/fire-portal.git
cd fire-portal
(cd backend && npm ci)
(cd frontend && npm ci)

# 2. configure env
cp backend/.env.example backend/.env
# edit backend/.env - set DATABASE_URL to your Neon URL
# (or a local Postgres) and set SESSION_SECRET to any random string

# 3. push schema + seed
cd backend
npx prisma generate
npx prisma db push --accept-data-loss
npx tsx prisma/seed.ts

# 4. run both processes (in two terminals)
cd backend && npm run dev           # http://localhost:4000
cd frontend && npm run dev          # http://localhost:5173
```

Log in with `admin` / `Admin@123` for a super-admin session.

10. Where to look when...

I want to...	Look here
Add a new API endpoint	<code>backend/src/routes/</code> , <code>controllers/</code> , <code>services/</code>
Add a new page	<code>frontend/src/pages/YourPage.tsx</code> + register in <code>App.tsx</code>
Change what a role can do	<code>backend/src/lib/rbac.ts</code> (single source of truth)
Change the theme colour	<code>frontend/tailwind.config.js</code> (<code>brand.*</code> scale)
Add a new report	<code>services/reportsService.ts</code> + <code>controllers/reportsController.ts</code> + <code>routes/reports.ts</code> + a new page in <code>pages/reports/</code>
Add a new equipment field	Prisma schema → migrate → controller Zod → service → frontend types → forms
Change frequency of scheduled inspections	<code>EquipmentType.inspectionFrequencyDays</code> (per-type) or <code>ChecklistTemplate.frequencyDays</code> (per-template override)
Add a new question type	<code>backend/src/lib/checklistTypes.ts</code> (the enum) + rendering in <code>InspectionPerformPage</code>

11. Things to improve for a full production rollout

- Object storage for photos** — replace `multer.diskStorage()` with S3 / R2 client. About half a day of work.
- Email / SMS / WhatsApp notifications** — pluggable in `services/notificationService.ts` (stub file exists). Recommend a provider like AWS SES + Twilio.
- Multi-instance HA** — the app is stateless; needs shared session storage (Redis) if scaled beyond one process.
- Enterprise SSO** — swap the password auth for SAML / OIDC in `services/authService.ts`. Roles still map to internal RBAC.
- Audit log retention** — currently indefinite; add a monthly cleanup job or archive to cold storage.
- CI pipeline** — GitHub Actions running `tsc` + tests + Prisma migrate check on every PR.

12. Further reading

- `ARCHITECTURE.md` — original architecture notes
- `API.md` — full endpoint catalogue

- `DATABASE.md` — schema deep-dive
- `RBAC.md` — role/permission matrix
- `DEPLOYMENT.md` — Render + Neon step-by-step
- `USER_MANUAL.md` — role-based end-user manual

Questions? The commit log tells the whole story of how each feature was built — `git log --oneline` is a great starting point.